

GRAFENWOHR, Germany

WMO: 106870

Lat: 49.699N

Lon: 11.940E

Elev: 415

StdP: 96.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-9.8	-16.0	1.0	-12.3	-12.6	1.3	-7.9	8.2	3.8	7.2	2.9	0.7	320	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	31.1	19.3	28.9	18.7	26.9	18.3	20.8	28.0	19.7	26.6	18.9	25.1	2.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.2	22.5	17.7	13.4	21.5	16.8	12.6	20.9	61.6	28.1	58.3	26.8	55.3	25.0	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.7	5.0	-16.7	34.4	3.9	1.8	-19.6	35.6	-21.9	36.6	-24.1	37.6	-26.9	38.9		
(5)			-17.4	22.7	3.7	1.3	-20.1	23.6	-22.3	24.4	-24.3	25.2	-27.0	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-1.0	0.4	3.8	8.2	12.6	16.1	18.3	17.5	12.9	8.3	3.7	0.8	(6)
(7)		DBStd	7.65	4.60	4.53	3.67	4.06	3.79	3.67	3.31	3.25	3.33	3.73	3.49	3.83	(7)
(8)		HDD10.0	1463	341	268	194	81	17	1	0	0	9	78	191	284	(8)
(9)		HDD18.3	3684	599	501	452	304	182	88	43	56	166	312	439	543	(9)
(10)		CDD10.0	917	0	0	1	27	97	183	257	231	96	25	1	0	(10)
(11)		CDD18.3	96	0	0	0	0	4	20	42	28	2	0	0	0	(11)
(12)		CDH23.3	1462	0	0	0	14	81	314	568	426	59	1	0	0	(12)
(13)		CDH26.7	462	0	0	0	1	12	99	206	135	9	0	0	0	(13)
(14)	Wind	WSAvg	1.8	2.1	1.9	2.3	1.9	1.7	1.7	1.4	1.3	1.4	1.8	1.9	(14)	
(15)	Precipitation	PrecAvg	769	66	54	53	42	69	78	81	69	64	58	66	70	(15)
(16)		PrecMax	1125	152	140	141	97	154	136	129	165	157	201	151	185	(16)
(17)		PrecMin	527	8	9	10	8	18	19	19	20	18	15	0	13	(17)
(18)		PrecStd	123	34	33	34	21	35	31	31	34	29	36	34	38	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	13.2	18.3	25.1	28.2	32.8	34.0	33.3	28.0	22.0	14.6	10.8	(19)
(20)			MCWB	8.8	8.3	11.1	14.2	18.4	20.2	19.8	19.7	17.7	14.9	11.7	8.9	(20)
(21)		2%	DB	8.4	10.8	15.8	22.1	25.8	29.8	31.2	30.1	25.0	18.8	12.2	8.9	(21)
(22)			MCWB	7.0	7.9	8.9	13.0	17.4	19.1	19.2	19.1	17.0	14.1	10.3	7.8	(22)
(23)		5%	DB	6.9	8.9	12.8	20.0	23.2	27.0	29.0	27.8	22.2	16.2	10.8	7.2	(23)
(24)			MCWB	5.7	6.4	7.9	12.3	15.3	18.3	18.6	18.9	15.8	13.0	9.3	6.3	(24)
(25)		10%	DB	5.1	7.0	10.1	17.1	21.0	24.5	26.5	25.7	20.0	14.2	9.0	6.0	(25)
(26)			MCWB	4.2	5.4	6.9	10.7	14.3	17.1	18.0	17.8	14.8	11.9	8.1	5.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.9	11.9	15.4	19.9	21.9	21.5	22.2	19.2	16.1	12.3	9.2	(27)
(28)			MCDB	10.1	12.1	16.0	22.5	25.9	29.1	28.4	27.4	24.4	19.4	13.4	10.0	(28)
(29)		2%	WB	7.2	8.1	9.9	14.1	18.1	20.2	20.3	20.6	17.7	14.8	10.7	7.9	(29)
(30)			MCDB	8.2	10.0	13.6	20.5	24.2	27.1	28.0	27.4	23.5	17.6	11.9	8.7	(30)
(31)		5%	WB	5.7	6.7	8.6	12.7	16.4	18.8	19.6	19.6	16.6	13.6	9.4	6.3	(31)
(32)			MCDB	6.4	8.3	11.8	18.8	21.7	25.2	26.6	26.3	21.4	16.0	10.5	6.8	(32)
(33)		10%	WB	4.1	5.3	7.3	11.4	14.9	17.6	18.7	18.6	15.5	12.3	8.1	5.4	(33)
(34)			MCDB	4.9	6.8	9.6	16.7	19.5	23.4	24.9	24.4	19.3	14.2	8.9	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	7.8	9.4	12.5	12.2	12.3	12.8	12.9	11.6	9.6	6.1	5.0	(35)
(36)			MCDBR	6.8	11.6	15.8	18.4	16.8	17.5	18.2	17.5	15.8	13.0	9.1	6.4	(36)
(37)			MCWBR	5.7	8.4	10.0	10.0	8.7	8.1	7.1	7.7	8.6	8.8	7.3	5.4	(37)
(38)		5% WB	MCDBR	6.2	9.8	13.4	16.6	14.8	15.7	15.0	15.2	14.2	12.2	8.4	6.2	(38)
(39)			MCWBR	5.6	7.7	9.1	9.5	8.4	7.8	6.7	8.0	8.1	8.5	7.2	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.320	0.358	0.393	0.399	0.399	0.410	0.403	0.374	0.349	0.324	0.295	(40)	
(41)		taud	2.412	2.365	2.329	2.270	2.285	2.310	2.276	2.316	2.375	2.436	2.460	2.427	(41)	
(42)		Ebn at Noon	753	821	845	850	860	861	843	831	819	777	715	703	(42)	
(43)		Edh at Noon	63	84	104	123	126	124	126	116	99	78	61	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.85	1.68	2.73	4.20	4.95	5.45	5.17	4.52	3.26	1.94	0.93	0.66	(44)	
(45)		RadStd	0.15	0.27	0.36	0.55	0.57	0.46	0.52	0.45	0.44	0.24	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	+0.30	+0.38	N/A	N/A	N/A	+20

Nomenclature: See separate page